

Projeto Lego Mindstorms EV3

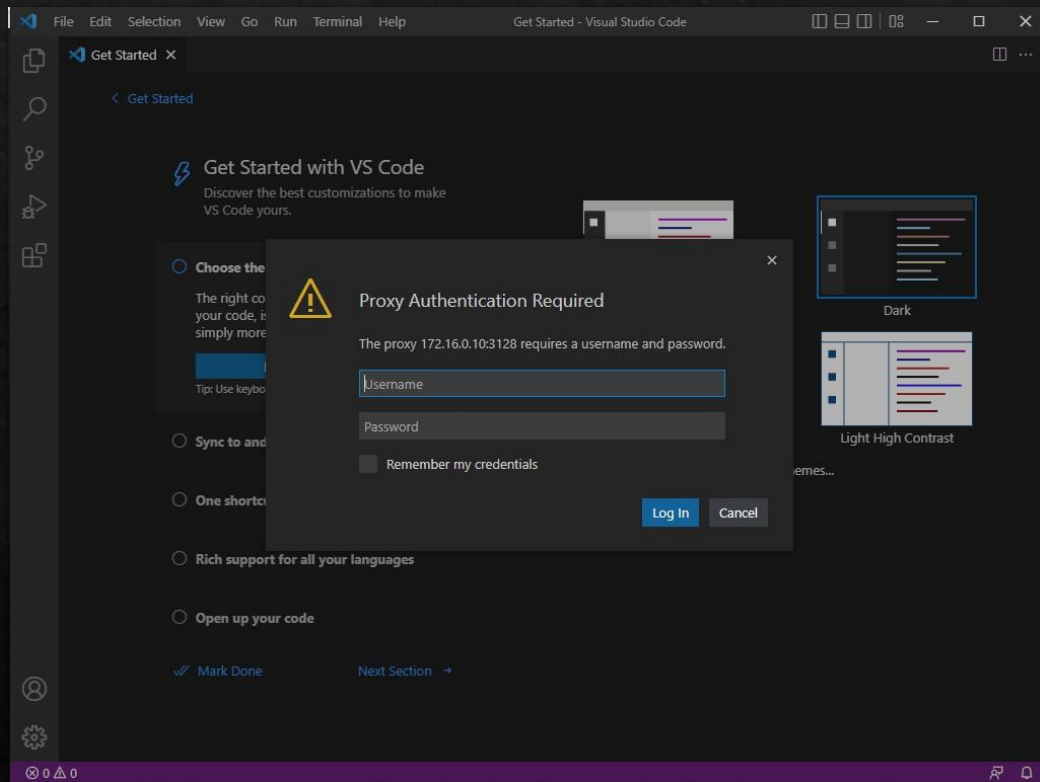
A dark, grayscale image of a Lego Mindstorms EV3 robot with a custom-built structure, serving as a background for the title. The robot is positioned in the lower half of the frame, with its EV3 brick at the top and a custom-built structure made of various colored Lego bricks (red, green, yellow, blue) extending downwards. The robot has two large black wheels and a smaller front wheel. The background is a solid dark gray.

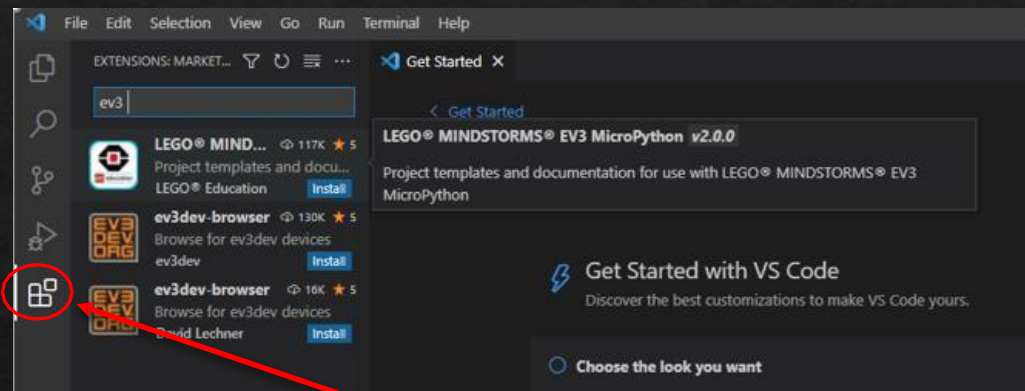
Ambiente Usado



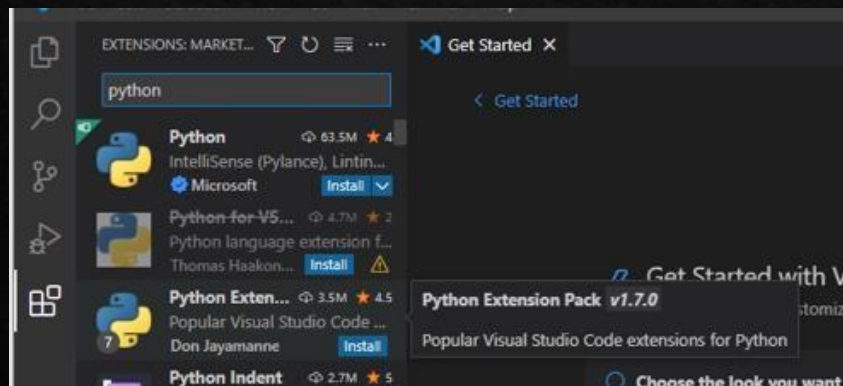
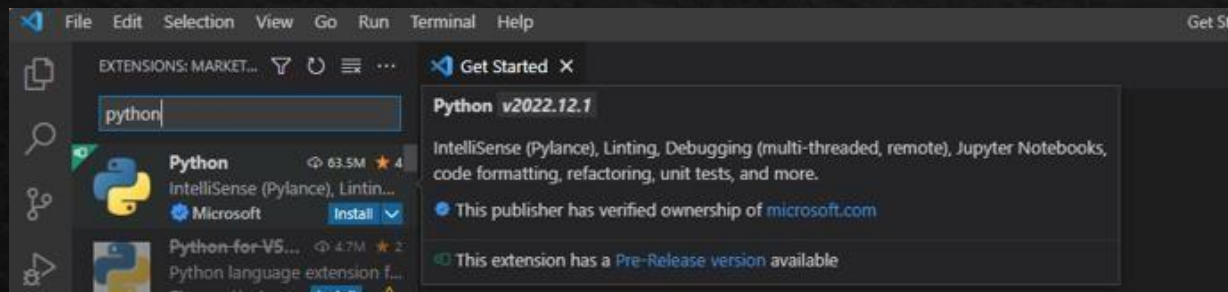
- ◊ Usaremos o Visual Studio Code para programar os robôs Lego EV3
- ◊ Nele há uma biblioteca que possibilita a comunicação do robô EV3 com o Visual Studio em Python

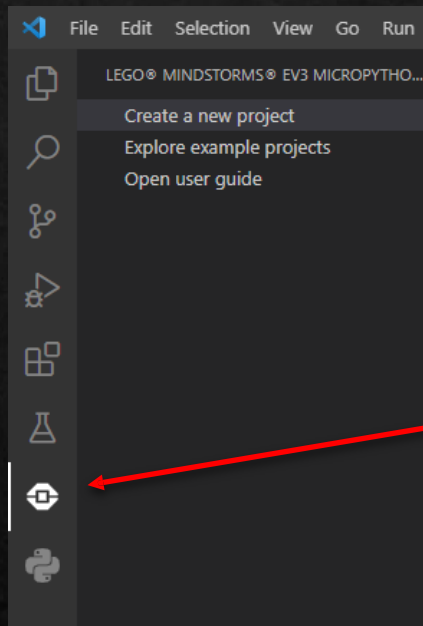
Usando a senha de login da internet insira seu login e senha para começar a instalar as bibliotecas caso não esteja instalado.



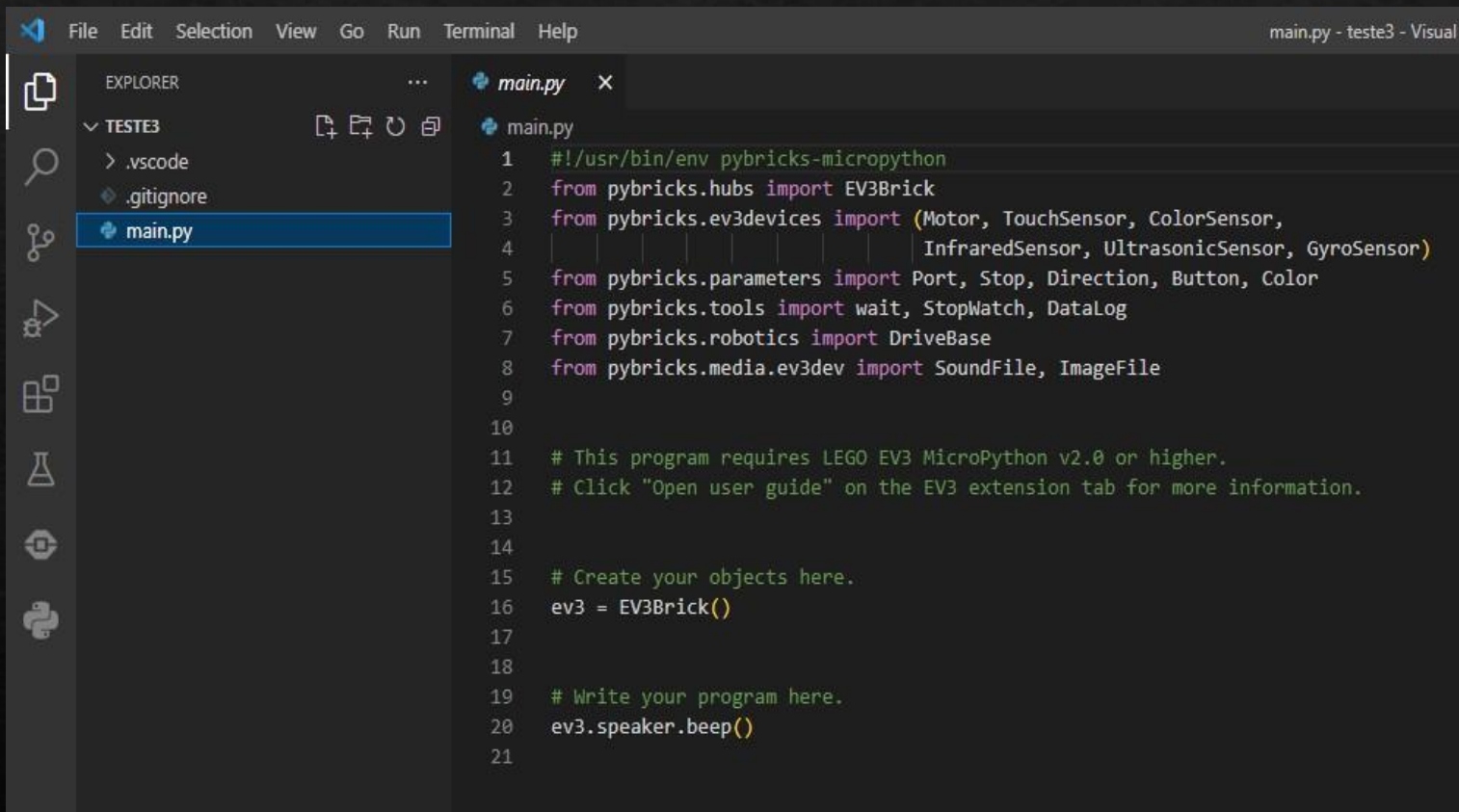


◆ Na aba de extensões procure pelas bibliotecas LEGO MINDSTORMS EV3 MicroPython, Python e a Python Extension Pack.



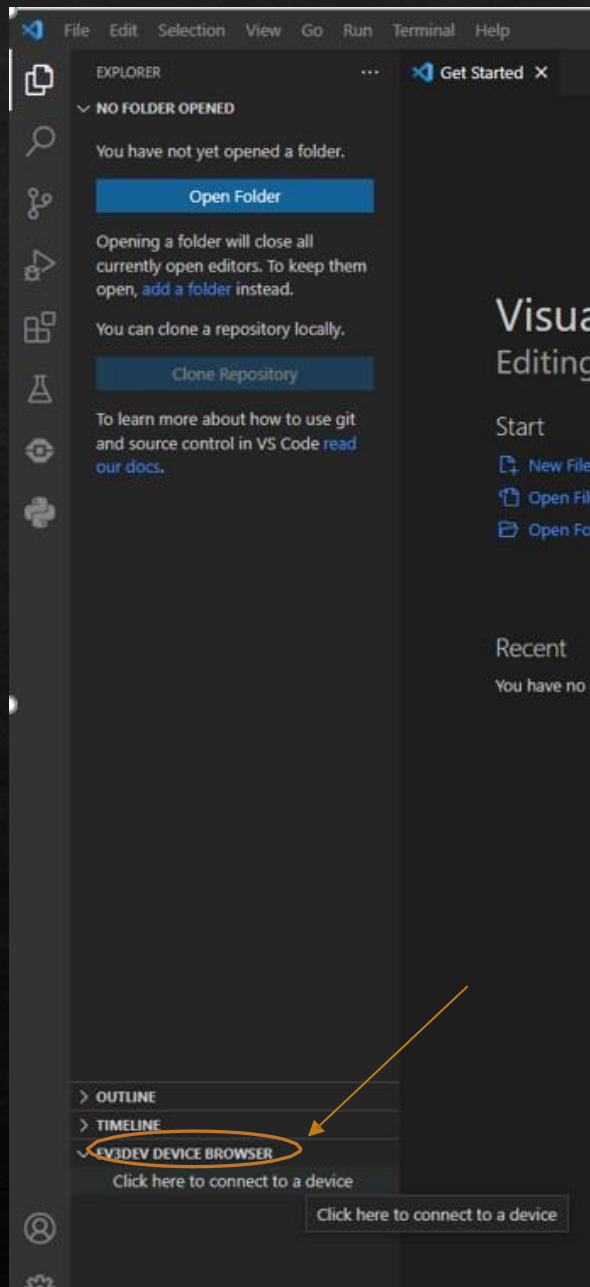


Na aba do LEGO MINDSTORMS crie um novo projeto,
nomeie-o, selecione um local para a pasta

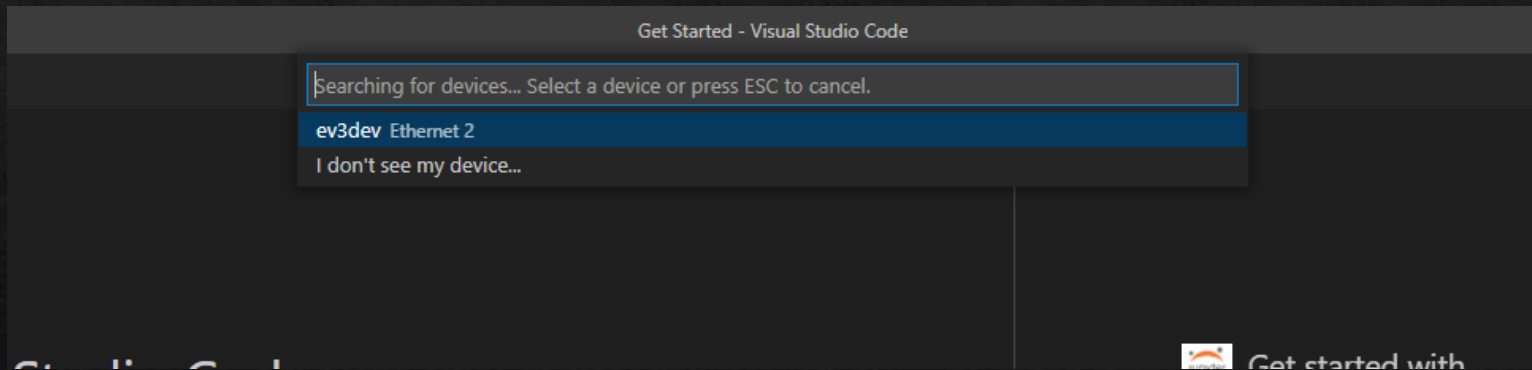


```
1  #!/usr/bin/env pybricks-micropython
2  from pybricks.hubs import EV3Brick
3  from pybricks.ev3devices import (Motor, TouchSensor, ColorSensor,
4  | | | | | InfraredSensor, UltrasonicSensor, GyroSensor)
5  from pybricks.parameters import Port, Stop, Direction, Button, Color
6  from pybricks.tools import wait, StopWatch, DataLog
7  from pybricks.robotics import DriveBase
8  from pybricks.media.ev3dev import SoundFile, ImageFile
9
10
11  # This program requires LEGO EV3 MicroPython v2.0 or higher.
12  # Click "Open user guide" on the EV3 extension tab for more information.
13
14
15  # Create your objects here.
16  ev3 = EV3Brick()
17
18
19  # Write your program here.
20  ev3.speaker.beep()
21
```

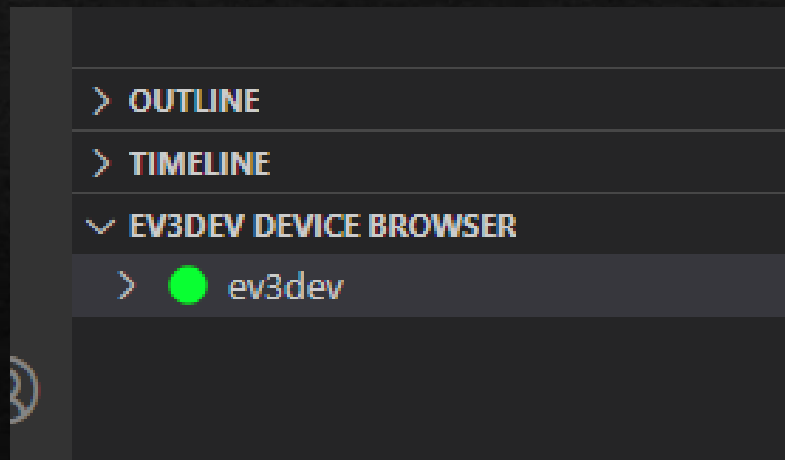
- ◆ Após criar a pasta, será criado automaticamente um código exemplo.
- ◆ Em seguida, será mostrado um código exemplo



◆ Na aba “Explorer” vá em EV3DEV
DEVICE BROWSER e conecte seu
EV3 **através da porta USB.**



◆ Escolha seu EV3 e espere conectar



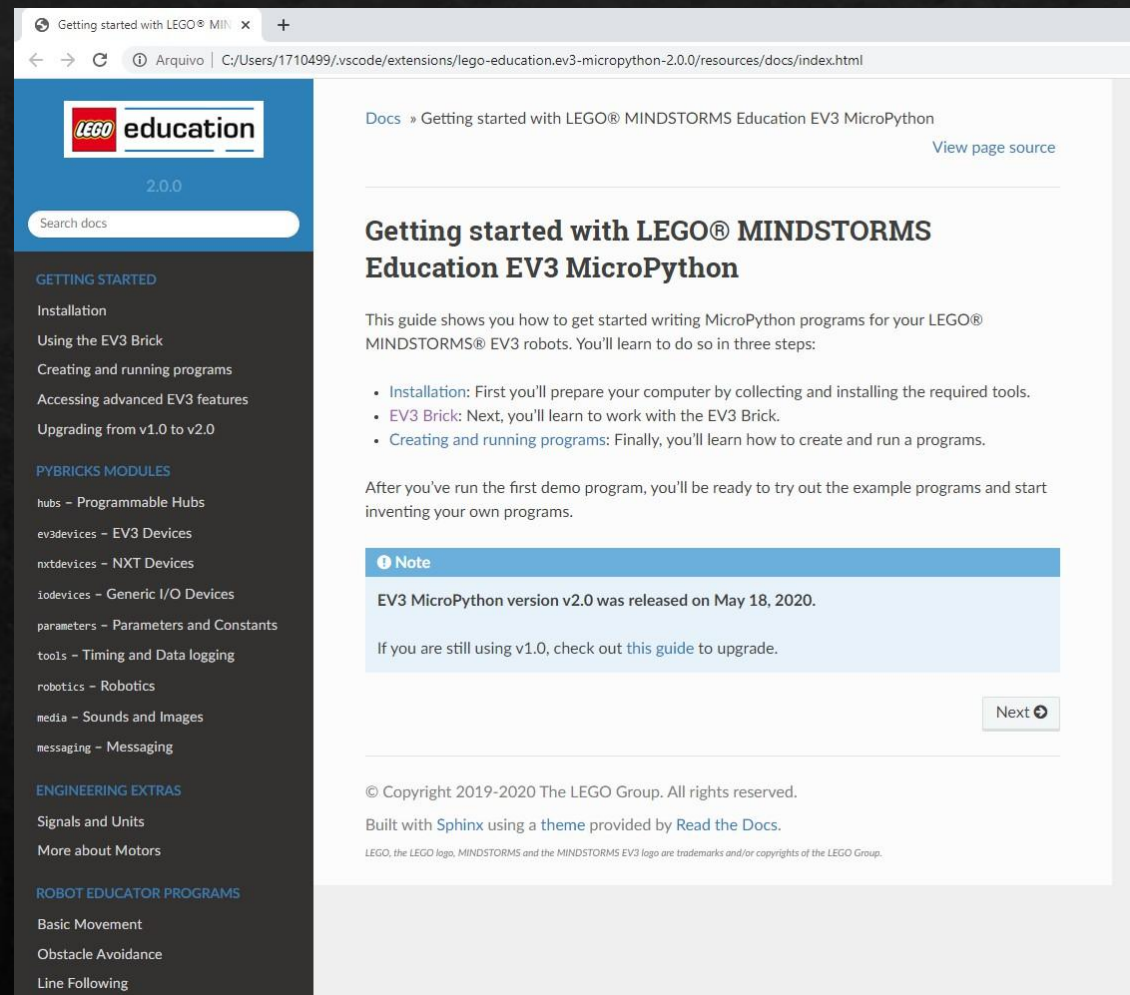
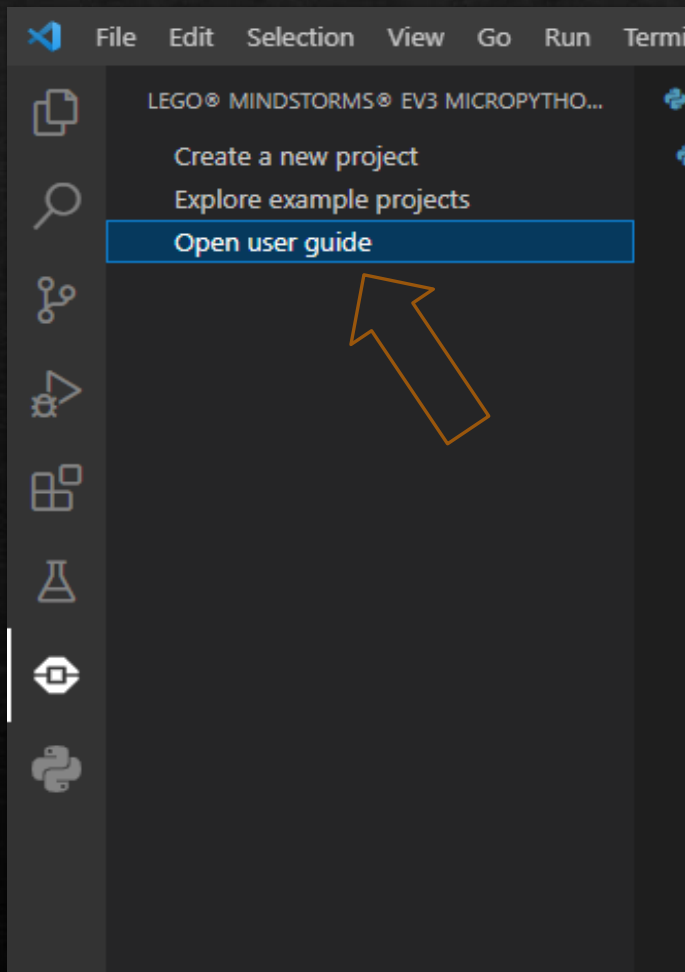
◆ Após conectar, aparecerá uma bolinha verde no canto inferior esquerdo indicando que está conectado

Sensores e Atuadores do Mindstorm EV3



Servo Motor médio

Servo Motor grande



O guia será a principal ferramenta de estudos de vocês!!!

Boa sorte!!!